Web Development Frameworks
(ITUE203)

BUILDING A DYNAMIC WEBSITE USING PHP AND MySQL

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1 Introduction

A dynamic website generates or updates content based on user input and data. PHP runs on the server to process requests, while MySQL stores and retrieves information from a database.

2 Problem Statement

Static websites cannot independently process user registration, save submitted information, or retrieve personalized records from a database. A PHP and MySQL application solves these limitations by enabling dynamic content and data management.

3 System Architecture

Frontend (HTML & CSS)
- User Interface
- Forms and Layouts

Web Server (Apache)
- Handles HTTP requests
- Executes PHP files

Backend (PHP)
- Business Logic
- Form Validation
- Database Connection

Database (MySQL)
- Stores User Data
- Executes SQL Queries

4 Working Process

1. User fills in a registration form
2. Browser sends form data to PHP
3. PHP validates the submitted data
4. PHP communicates with MySQL
5. MySQL stores the information
6. PHP returns success or error response

5 Code Example

1. Database Connection (db_connect.php)

```php
<?php
$host = "localhost";
$user = "root";
$pass = "";      // your password
$db   = "studenthub";

$conn = new mysqli($host, $user, $pass, $db);

if ($conn->connect_error) {
   die("Connection failed: " . $conn->connect_error);
}
?>
```

2. Insert Data Using Prepared Statement (register.php)

```php
<?php
include 'db_connect.php";

$name  = $_POST['name'];
$email = $_POST['email'];
$pass  = password_hash($_POST['password'], PASSWORD_DEFAULT);

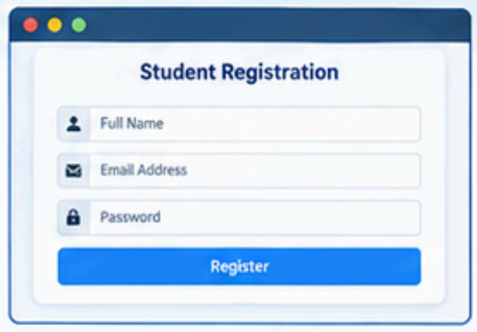
$stmt  = $conn->prepare("INSERT INTO students
                         (name, email, password)
                         VALUES (?, ?, ?)");
$stmt->bind_param("sss", $name, $email, $pass);

if ($stmt->execute()) {
    echo "Registration successful!";
} else {
    echo "Error: " . $stmt->error;
}
$stmt->close();
$conn->close();
?>
```

6 Database Design (Example)

Table: students

Field	Type	Description
id	INT (AUTO_INCREMENT)	Primary Key
name	VARCHAR(100)	Student Name
email	VARCHAR(100)	Student Email
password	VARCHAR(255)	Hashed Password
created_at	TIMESTAMP	Registration Date



Example Registration Form

7 Real-World Application

- **Student Management Portal** — Register, login, view profile, manage records
- **Library Management System** — Track books, members, issues/returns
- **Online Registration System** — Events, courses, examinations
- **Inventory Management** — Product tracking and stock updates

8 Advantages

- Easy to learn and use
- Open source and cost effective
- Supports dynamic web applications
- Large community support
- Works well with MySQL databases

9 Security Considerations

- Use prepared statements to prevent SQL injection
- Validate and sanitize user input
- Hash passwords (e.g., *password_hash()*)
- Use sessions and cookies securely
- Implement proper error handling

10 Conclusion

PHP and MySQL together provide a simple yet powerful way to build dynamic websites. They allow us to process user requests, manage data, implement security measures and develop real-world web applications such as student portals, library systems and online registration platforms.